

FOSSIL FUEL EFFECTS ON THE GLOBAL CARBON CYCLE

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INTRODUCTION

The global carbon cycle is the intricate process by which carbon circulates among the Earth's principal reservoirs: the atmosphere, oceans, biosphere, and geosphere. As a critical biogeochemical cycle, it governs the movement of carbon in various forms, enabling the regulation of life and climate across the planet. Modelling such a complex cycle in its true form is quite the task. For calculation and analysis, the cycle was condensed into a rudimentary box-model structure.

Box models are a simplified way to represent complex environmental systems. They break down the system into different "boxes" or reservoirs and track the flow of carbon between these reservoirs. Each box stores carbon, while arrows represent the rate of transfer (flux) between them.

The flux values used in this study have been set to output steady-state values, i.e., net flux of carbon is equal to zero. In general, this would imply that the rate of change of the box contents can be boiled down to a sum of its influxes and outfluxes. This is helpful as it transforms these calculations into an ODE, which can then be solved using the Runge-Kutta and Euler methods. Box models typically use ODEs to represent these flows, and Runge-Kutta provides a precise way to solve them. We use these methods for analysing how carbon levels in various reservoirs evolve under different scenarios, such as increased deforestation or efforts to mitigate carbon emissions.

In addition to this, we used data from the Byrne et al. (2024) paper in Nature entitled "Carbon emissions from the 2023 Canadian wildfires" and added further forcings in for different forest fire scenarios to see the model response to that. The data used was based on the carbon emitted by the 2023 forest fires, and we used the same rate of forest land cover loss to see how the model responds to that, giving us a view of what could be a probable future outcome.

METHODS AND RESULTS

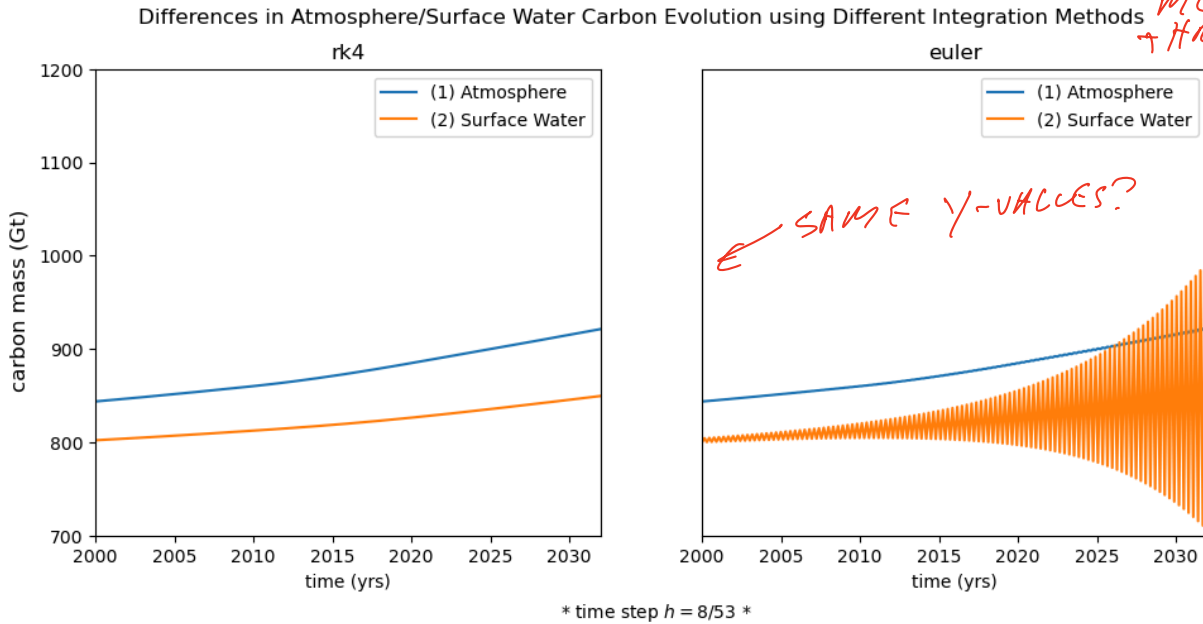
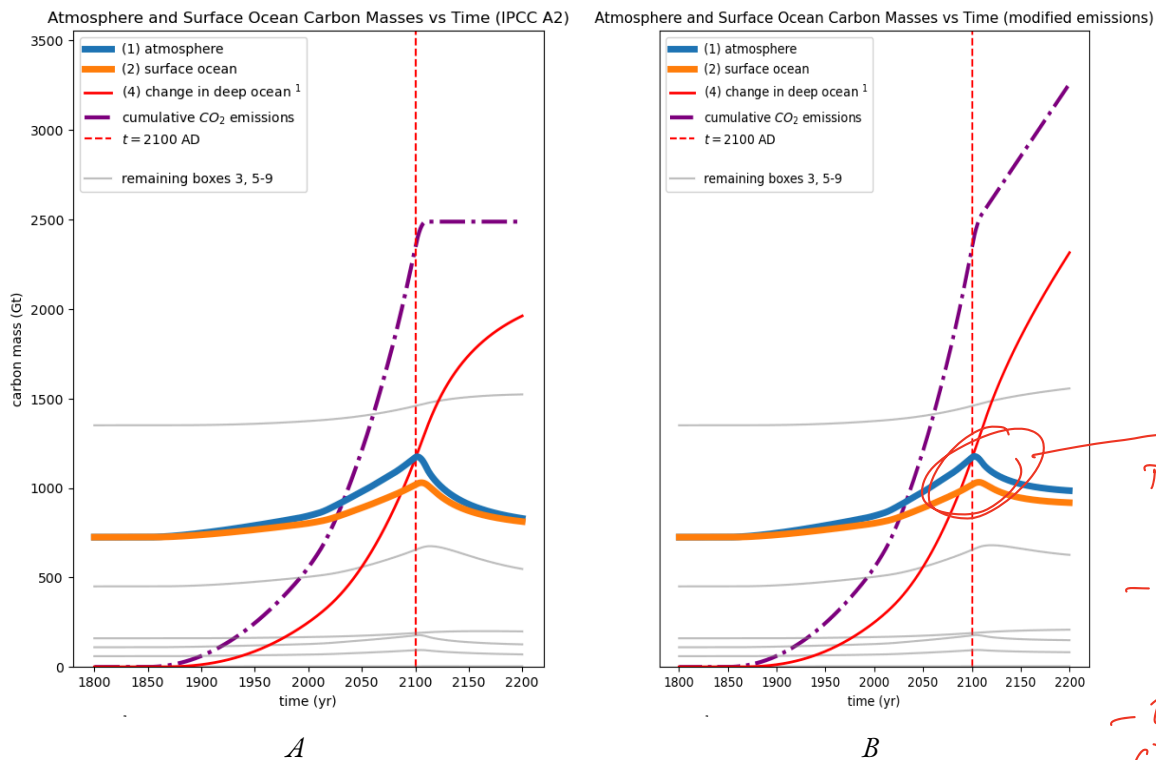


Figure 1 demonstrates the difference in RK4 and Euler techniques when the time step is equal to 2 times the reciprocal of the biggest rate constant ($2/12$). The time step taken is approximately 0.1666. This is a zoomed in view in order to point out the Euler explosion. ?



¹ box (4)'s mass values are subtracted by $m_0 = 37675$

NICE IDEA
BUT WHY DO THAT
AND WHAT DID
YOU LEARN?

Figure 2 plots the IPCC A2 model with respect to the surface ocean and atmospheric masses from the box model. Figure 2A is the emissions data plotted as is and B is the emission data modified to have the emission value after 2100 AD to be equal to that from 2000 AD.

THIS IS THE FIRST ARTICULATION THAT YOU ARE EVEN DOING THIS?

? BUT YOU ARE FORCING THE ATMOSPHERE?

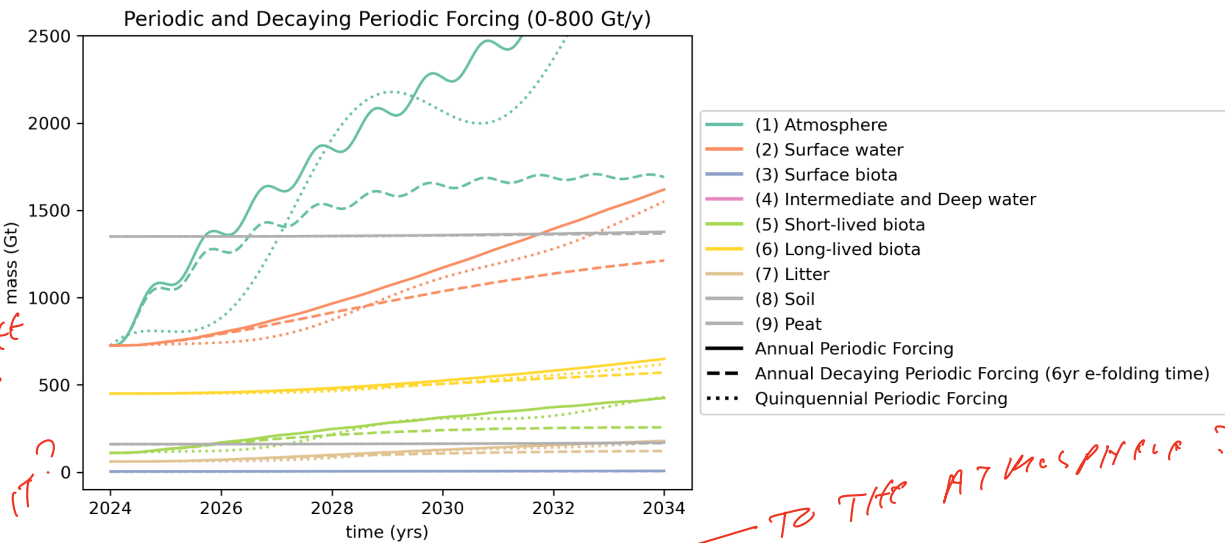


Figure 3 depicts the annual and quinquennial forcing in addition to a decaying period forcing.

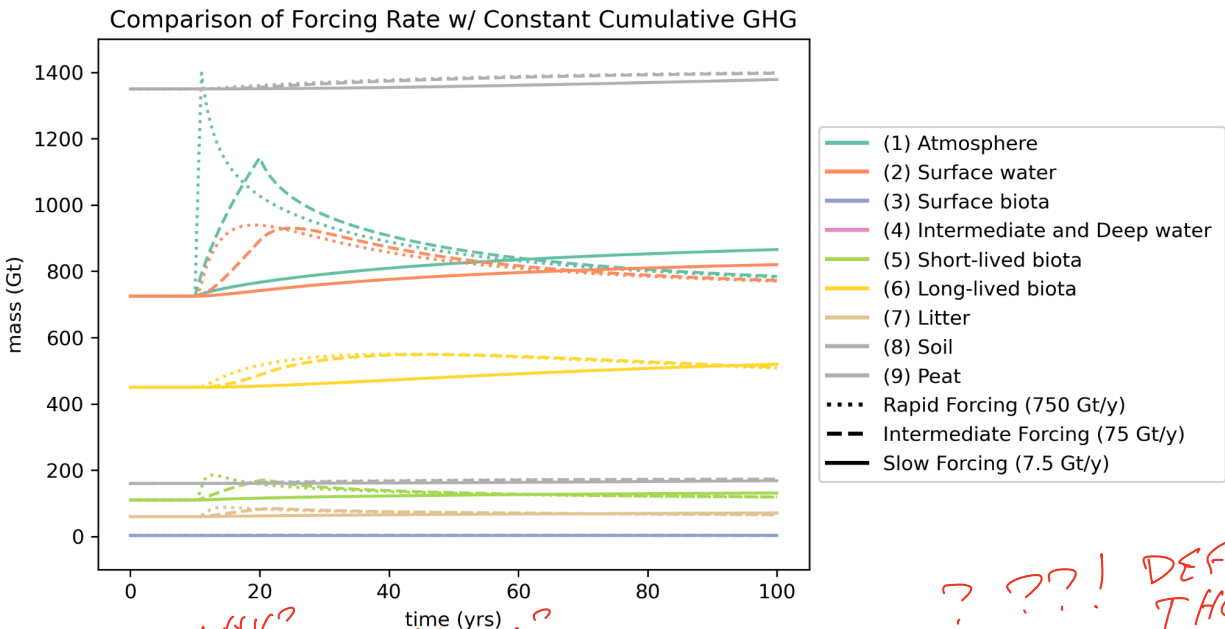


Figure 4 750 Gt of C is added to the model as a source. The addition is done at 3 speeds: Rapid, Intermediate and Slow. It can be observed that the increase is immediate and short-lived for the rapid forcing, and gets more gradual when the forcing time changes.

WAGUE

WAGUE

? ??! DEFINE THESE RATES PLEASE
YOUR CAPTIONS SHOULD BE MOSTLY NARRATING.

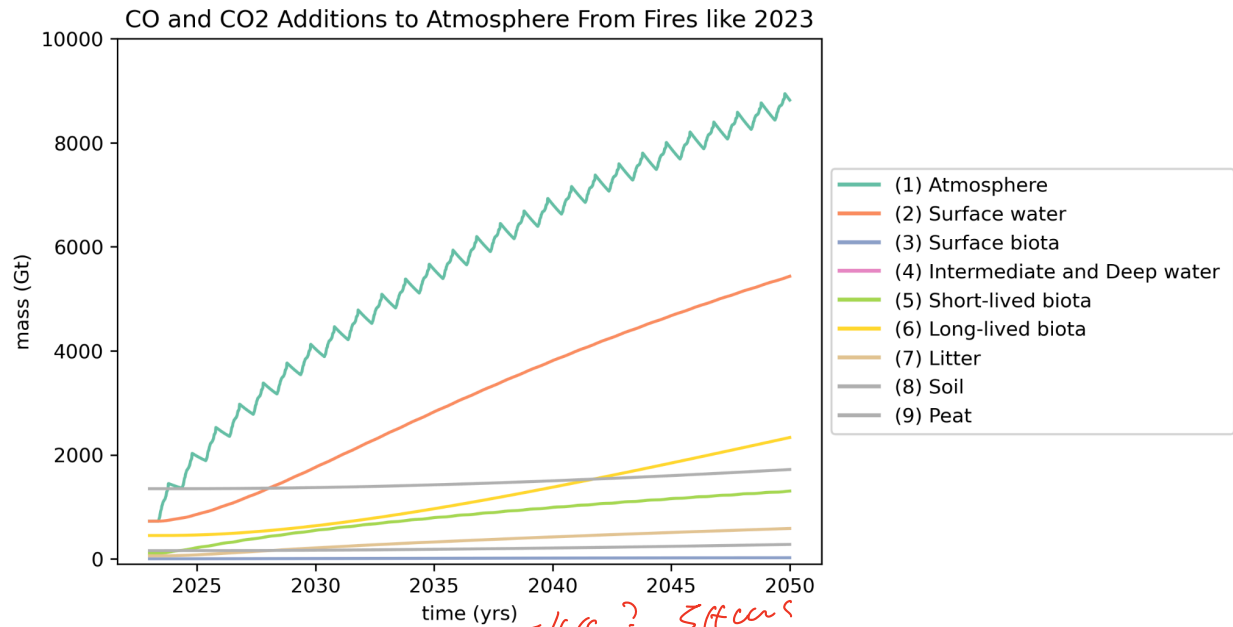


Figure 5 shows a case where the emissions from the 2023 fires as seen in Byrne et al. (2024) are replicated to happen every year. (This shows a stark rise in flux into the atmosphere and the biosphere every year.

How ABOUT
'REPEAT
'YEARLY"
DURING SUMMER

WHEN?

??
GIVE THE
VALUE

WHEAS WHAT?
YOU HAVE 5 BOXES
(BIOTA + PEAT) THAT
ARE PART OF THE
BIOSPHERE. WHAT
DO YOU MEAN?

RESULTS

→ PARAGRAPH BY PARAGRAPH PRESENTATION
OF WHAT IS ON EACH FIGURE

THE DISCUSSION IS VERY HARD TO FOLLOW.
→ YOU ARE INTERWEAVING ACTUAL
RESULTS AND MUSINGS AND
DOING IT ALL WITHOUT PRECISION.

DISCUSSION

Figure 1 shows a distinction between the 2 used techniques when the time step is altered. If the time step taken is larger than twice the reciprocal of the biggest rate constant (12), the Euler method would become inconsistent and explode. Here, we take a time step (0.15), slightly smaller than $2/12$ (0.1666), and notice the Euler method stops functioning. 0.1666 works as the Nyquist frequency. However, the Runge-Kutta plot doesn't suffer the same fate as it is more precise. The RK method uses 4 distinct slopes for the time step, providing more accuracy and stability than Euler and thus doesn't explode when the time step gets closer to the Nyquist. It is also worth noting that RK4 takes 4 times longer to compute than Euler, thus a tradeoff between efficiency and accuracy is observed.

This motivated us to use RK4 for modelling the nine-box model over time. Next, we took data from the emissions.py document which is based on the 2012 IPCC report for CO₂ emissions. We iterated this over time for the time scale of our choice and plotted **Figure 2**. This figure depicts the box 1 and box 2 masses alongside the cumulative CO₂ emissions. Note that the difference between the sum of all nine boxes' final values and the sum of all nine boxes' initial values equates to the final cumulative CO₂ emission value, validating our plots and values. **Figure 2 A** is the emission model as is, i.e., the value for emissions after 2100 AD goes to 0, flattening the curve. However, **Figure 2 B** is modified to give the 2000 AD value to the years after 2100, which was 8.125 Gt/year. The difference in the CO₂ values is significant, with the curve going upwards. This shows that even with values from 2000 AD, the emission rate is alarming. A key point to note in this figure is the line depicting the change in deep ocean mass. Despite the flux from the deep ocean being inconsequential when compared to its mass, we see a perceptible change in deep ocean mass. This implies that a majority of the CO₂ emitted eventually ends up in the deep ocean in the future.

Figure 3 involves the forcing being made periodic. This did not have any noteworthy changes in the plots for boxes 2-9. Focusing on box 3 (surface biota), no difference is observed at all. This is attributed to its high rate constant, which implies that it gives out carbon faster than it can hold on to it. This can be connected to the nature of surface biota to be interdependent with surface water environment and chemistry. The remaining boxes show no major changes between the annual and quinquennial periodic forcings, which could be because the time scale is so large. If the forcing were to be decennial or higher, we may have seen some changes.

Figure 4 is a new emission scenario added to the model. A forcing of 750 Gt was added into the model and the changes were noted. Of this, we pose 3 scenarios for the duration of forcing. Rapid forcing is when 750 Gt of C is added within a span of a year, intermediate is when it's added over 10 years (75 Gt/yr) and slow is when it's added in over 100 years (100 Gt/yr). The rapid forcing scenario sees a sudden spike in carbon flux, which is not sustained for very long. The slow scenario sees a more gradual rise, but its effects are observed for longer. The notable result of this experiment is that the rate of forcing has major impacts in the short term for the boxes closely connected to the atmosphere (ie. surface water) and very little impact on the boxes further away (ie. litter). In the long term all of the models converge to the same place.

Figure 5 is a case study where the data from Byrne et al. (2024) is applied to our pre-existing model. We modelled a scenario where fires akin to those in 2023 happen every summer. If this were to occur the atmosphere would be unable to move carbon into other locations at anywhere near the rate the fires force carbon into the atmosphere, the magnitude of the influx of CO₂ into the atmosphere would be too much for the atmosphere to handle.

THIS NARRATIVE PART IS BASICALLY IMPOSSIBLE TO FOLLOW.

- YOUR INTRO SHOULD LAY OUT YOUR QUESTIONS, EXPERIMENTAL PLAN AND YOUR GOAL. FROM NUMERICAL TECHNIQUES TO C-FORCING SCENARIOS
- YOUR RESULTS SHOULD STEP THROUGH THE EXPERIMENTS IN A NARRATIVE:
 - EACH EXPERIMENT, FIGURE BY FIGURE
 - EACH FIGURE $\Rightarrow$ 1 PARAGRAPH
 - SUMMARY OF KEY RESULTS AT END.
- YOUR DISCUSSION MUST EMERGE FROM SPECIFIC AND WELL-DEFINED RESULTS.

CONCLUSION $\rightarrow$ WHAT CONCLUSIONS DO YOUR SPECIFIC RESULTS REQUIRE YOU TO ARTICULATE?
 $\rightarrow$ WHAT IMPLICATIONS ARE SUPPORTED BY

CONCLUSION

INFERENCES DRAWN
FROM YOUR RESULTS.

Future directions for this project aim for increased accuracy. With more time and resources, the precision of the integrators used can be prioritised over efficiency. A deeper dive could be taken into how forest fire emissions work and how the carbon emitted is distributed to the other biotic and abiotic components. It would also be of interest to study all the boxes individually, and force the same amount of carbon in them to see how each box behaves on its own. A further study into converting the model to a grassland upon 20% loss of forest area would be intriguing. In all, the carbon cycle is a very complex process and more tools and models to understand its nuances better would benefit in a clearer picture of how it functions and how detrimental forest fires are.

IT LOOKS LIKE YOU DID SOME
NEAT EXPERIMENTS. HOWEVER, THIS
PAPER IS DIFFICULT OR IMPOSSIBLE
TO FOLLOW.

C/C⁺
IF YOU WOULD LIKE TO JUSTLY IMPROVE
THIS MARK, PLEASE REWRITE THIS
PAPER BY NOV. 18.

⇒ PLEASE SEE THE GUIDELINES
ON THE WEBSITE FOR YOUR
WRITING. YOU'VE ALSO NOW READ
A LOT OF PAPERS. ~~THE~~ ^{HOW} READ
YOU HAVE EXAMPLES OF METHODOLOGY,
SPECIFIC AND DIRECT PRESENTATION.

CONTRIBUTIONS

Ethan: Figures 1-2

Josh: Figures 3-5

Shreya: Report